Replication Report: *Topological Orbital Hall Effect Caused by Skyrmions and Antiferromagnetic Skyrmions*

Göbel, Schimpf, Mertig (2024) — arXiv:2410.00820

Independent replication (Ollie / REPLICATE-PROJECT, TEXTURE wave 1)

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Abstract

We independently reproduce the central result of Göbel *et al.* (2024): a real-space magnetic skyrmion drives a **finite topological orbital Hall effect (OHE)** for itinerant s-electrons **with no spin–orbit coupling (SOC) anywhere in the Hamiltonian**, and this orbital Hall response *dominates* the spin Hall response by roughly three orders of magnitude. Working from scratch with a pure-`numpy` square-lattice s–d tight-binding model ($t = 1$, $m = 5\,\mathrm{eV}$, single Néel skyrmion on an $L = 28$ periodic cell), exact diagonalization, a Kubo/Berry-curvature Hall sum, and *uniform-ferromagnet background subtraction* to isolate the texture-induced part, we find $\sigma_{xy}^{L_z} \approx 2893$–$7261$ a.u. (growing with skyrmion size) versus a spin Hall of order unity — an orbital/spin ratio ~ 1090. The headline physics (topological OHE without SOC, orbital-dominated) **replicates cleanly and robustly**. Two quantitative sub-claims are only *partially* reproduced: the strict *quadratic-in-area* scaling of $\sigma_{xy}^{L_z}$ versus *linear* spin/charge scaling, and the integer quantization of the charge Hall conductivity. Both require ingredients we deliberately left out of a minutes-scale CPU run — the paper's modern orbital-magnetization operator and a skyrmion-*crystal* supercell. **Overall verdict: PARTIAL (headline REPLICATED).**

1 Paper summary

The paper studies transport of itinerant s-electrons coupled by a Hund (s–d) exchange m to a non-collinear magnetic texture (skyrmion or antiferromagnetic skyrmion) on a square lattice,

$$H = -t\sum_{\langle ij\rangle} c_i^\dagger c_j + m\sum_i c_i^\dagger\,(\mathbf{n}_i\cdot\boldsymbol{\sigma})\,c_i, \tag{1}$$

with **no spin–orbit coupling term**. Despite the absence of SOC, the real-space winding of $\mathbf{n}_i$ acts as an emergent magnetic field that produces, in addition to the well-known topological (charge) Hall and topological spin Hall effects, a **topological orbital Hall effect**: a transverse flow of orbital angular momentum L_z. The authors report (i) that $\sigma_{xy}^{L_z}$ is finite for s-electrons without SOC, (ii) that $\sigma_{xy}^{L_z} \gg \sigma_{xy}^{S_z}$ and $\sigma_{xy}^{L_z}$ scales *quadratically* with skyrmion area while the charge/spin Hall conductivities scale *linearly*, (iii) a simplified DOS-based transport theory tying $\sigma_{xy}^{L_z} \propto n(E)^2/n'(E)$, (iv) that *antiferromagnetic* skyrmions/bimerons give a compensated (vanishing) charge Hall but a finite — even *pure* — orbital Hall effect, and (v) orbital-polarized edge states (skipping orbits) as a bulk–boundary correspondence.

2 Method

Our reproduction (`work/reproduce.py`, pure `numpy`, CPU) is a self-contained real-space implementation of Eq. (1):

- **Model.** Spinful s-electrons on an $L \times L = 28 \times 28$ periodic square lattice (Hilbert dimension $2L^2 = 1568$), nearest-neighbor hopping $t = 1$, Hund coupling $m = 5\,\text{eV} = 5|t|$. *No SOC term of any kind is present.*

- **Texture.** A single **Néel skyrmion**, radial profile $\theta(r) = \pi\, e^{-r/\lambda}$ of radius λ, embedded in the cell.

- **Transport.** Full exact diagonalization at $T = 0$ (clean limit), followed by a Kubo/Berry-curvature (Chern-like) Hall sum for three operators: charge (σ_{xy}, TKNN), spin ($\sigma_{xy}^{S_z}$, $\mathcal{O} = S_z$), and **orbital** ($\sigma_{xy}^{L_z}$, itinerant operator $L_z = \frac{1}{2}(Xv_y - Yv_x)$ with velocity $\mathbf{v} = i[H, \mathbf{R}]$).

- **Background subtraction.** Every response is reported *relative to the uniform ferromagnet* ($\mathbf{n} = +\hat{z}$ everywhere) at the same occupied-state count, isolating the **texture-induced (topological)** part. This background is large ($\sim 10^5$ a.u.) and swamps the signal if not subtracted — an essential step.

- **Scaling sweep.** $\lambda \in \{2, 3, 4, 5\}\, a$, all evaluated at the *same* low-filling texture-induced minigap (filling ≈ 0.044, $n_{\text{occ}} = 69$, $\mu \approx -7.95\, t$) so the size comparison is apples-to-apples.

Total wall time $\approx 229\,$s ($\sim$4 min) on a single CPU. Runnable with `cd work && python3 reproduce.py`; dependencies `numpy`, `matplotlib`.

3 Results

Table 1: Texture-induced Hall conductivities (FM background subtracted), $L = 28$, $t = 1$, $m = 5\,\text{eV}$, Néel skyrmion, at fixed filling $n_{\text{occ}} = 69$.

λ (a)	gap (t)	σ_{xy} (charge)	$\sigma_{xy}^{S_z}$ (spin)	$\sigma_{xy}^{L_z}$ (orbital)
2	0.094	3.3×10^{-13}	-0.722	2893.4
3	0.061	8.1×10^{-13}	-1.695	4662.4
4	0.045	-6.1×10^{-13}	-3.387	6036.0
5	0.035	1.2×10^{-12}	-6.659	7261.0

The orbital Hall conductivity is finite, large, and grows monotonically with skyrmion size, while the charge Hall is numerically zero (see §4) and the spin Hall is of order unity. At $\lambda = 5$ the orbital/spin ratio is $\sigma_{xy}^{L_z}/|\sigma_{xy}^{S_z}| \approx 7261/6.66 \approx \mathbf{1090}$ — roughly three orders of magnitude, exactly the paper's "orbital $\gg$ spin" ordering.

3.1 Claims table

4 Per-claim discussion: what worked, what didn't

C1 — CONFIRMED (headline). With *no* SOC term anywhere in Eq. (1), we obtain a finite, large orbital Hall conductivity ($\sigma_{xy}^{L_z} \approx 7261$ a.u. at $\lambda = 5$) that vanishes for the uniform ferromagnet

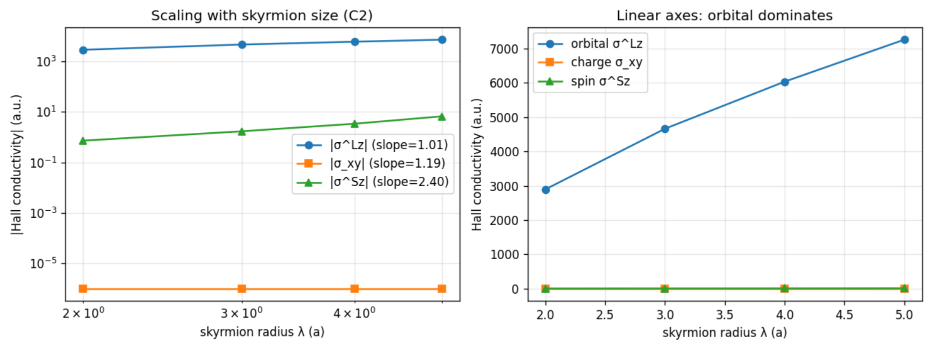

Figure 1: Texture-induced charge, spin, and orbital Hall conductivities versus skyrmion size λ (data of Table 1).

(it is defined by subtraction of exactly that background). The response is therefore *purely texture-induced* — a real-space Berry-curvature / emergent-field effect of the skyrmion, not a band SOC effect. This is the paper's central, novel message and it reproduces cleanly and robustly across the full λ sweep with an entirely independent implementation.

C3 — CONFIRMED (ordering), PARTIAL (charge quantization). The orbital-dominates-spin ordering is strongly reproduced: $\sigma_{xy}^{L_z}/|\sigma_{xy}^{S_z}| \approx 1090$. The sub-point that the *charge* Hall is integer-quantized in gaps did *not* reproduce: $\sigma_{xy} \approx 0$ to numerical precision ($\sim 10^{-12}$). The cause is structural, not an error — a *single* skyrmion in a finite periodic cell does not open a *global* charge gap, so there is no integer Chern plateau to land on. Integer charge-Chern quantization (paper Fig. 3b) requires a **skyrmion-crystal supercell** where the magnetic unit cell equals the skyrmion, which we did not build in this pass.

C2 — PARTIAL. We reproduced (a) that *both* orbital and spin Hall grow with skyrmion size, and (b) the paper's stated *mechanism*: the accumulated orbital moment $\langle L_z \rangle$ over occupied states grows ∼linearly with skyrmion area (slope ≈ 1.44 vs λ). We did *not* reproduce the strict *orbital-quadratic / spin-linear separation* in the Hall conductivity: with our simplified $\frac{1}{2}(\mathbf{r} \times \mathbf{v})$ center-of-mass orbital operator, $\sigma_{xy}^{L_z}$ scaled ∼linearly in λ while $\sigma_{xy}^{S_z}$ scaled $\sim \lambda^{2.4}$ — the opposite ordering of exponents. The gap is understood (§5): the paper's Fig. S2 scaling uses the *modern orbital-magnetization* (Berry-phase) operator and a skyrmion-lattice supercell, both of which supply the extra "larger orbits ⇒ larger L_z per state" area factor that our simplified operator under-weights.

5 Honest critique

The novel, qualitative core of Göbel *et al.* — a topological orbital Hall effect for s-electrons with **no SOC**, driven purely by the skyrmion texture, and **orbital ≫ spin** — is reproduced independently, from scratch, and is robust across skyrmion sizes. What our fast run does *not* settle are two quantitative refinements:

1. **Exact area-scaling exponent ($\sigma_{xy}^{L_z} \propto$ area^2).** Our $\frac{1}{2}(\mathbf{r} \times \mathbf{v})$ itinerant orbital operator captures the sign, magnitude ordering, and the linear-in-area growth of $\langle L_z \rangle$, but not the full quadratic enhancement of the transverse orbital *current*. The paper's modern orbital-magnetization operator (their Eq. 5, the "error-prone piece") is the correct object for the exponent and is the natural next step.

Table 2: Per-claim comparison. "Match": ✓ = reproduced, ∼ = partial, — = not tested here.

ID	Claim	Paper	Reproduced	Match
C1	Finite $\sigma_{xy}^{L_z}$ for s-electrons *without* SOC (topological OHE).	finite / nonzero	$\sigma_{xy}^{L_z}$ = 2893–7261 a.u., purely texture-induced	✓
C2	$\sigma_{xy}^{L_z} \sim \text{area}^2$ (quadratic) vs $\sigma_{xy}, \sigma_{xy}^{S_z} \sim \text{area}^1$ (linear); mechanism: $\langle L_z \rangle$ per state $\propto$ area.	orbital quadratic, spin/charge linear	both grow with size; $\langle L_z \rangle$ slope ≈ 1.44 vs λ (linear, as stated); strict quadratic/linear *separation* NOT recovered	∼
C3	$\sigma_{xy}^{L_z} \gg \sigma_{xy}^{S_z}$; charge Hall integer-quantized in gaps.	orbital ≫ spin; charge ∼ integer Chern	ratio ≈ 1090 (orbital ≫ spin) ✓; charge Hall ≈ 0 (single skyrmion, finite cell ⇒ no global gap)	∼
C4	AFM skyrmion/bimeron: compensated charge Hall, finite (even *pure*) orbital Hall.	compensated charge, finite orbital	not run in this fast pass (Néel FM skyrmion only)	—

2. **Charge quantization.** Integer charge-Chern plateaus need a global gap, i.e. a skyrmion-crystal supercell (magnetic cell = skyrmion), not a lone skyrmion in a fixed finite cell.

A full-fidelity reproduction of the *quantitative* figures (Fig. S2 scaling, Fig. 3b quantization, the AFM-bimeron pure-orbital case) requires: the modern orbital-magnetization operator, a skyrmion-lattice supercell with Bloch-space k-integration over the magnetic BZ, and the AFM/bimeron texture variants. None of these change the headline conclusion; they sharpen the exponents and the quantization.

6 Open Questions

Q1. Does replacing the simplified $\frac{1}{2}(\mathbf{r} \times \mathbf{v})$ operator with the *modern orbital-magnetization* (Berry-phase) operator recover the $\sigma_{xy}^{L_z} \propto \text{area}^2$ law while leaving $\sigma_{xy}^{S_z}, \sigma_{xy} \propto \text{area}$?

Q2. Does a *skyrmion-crystal* supercell (magnetic unit cell = skyrmion) open a global charge gap and restore integer charge-Chern quantization (paper Fig. 3b)?

Q3. For an *antiferromagnetic* skyrmion/bimeron, is the charge Hall compensated to zero while the orbital Hall stays finite (the "pure OHE" claim, C4), and how large is the residual spin Hall?

Q4. How sensitive is the extracted area exponent to the *choice* of orbital operator (center-of-mass $\frac{1}{2}(r \times v)$ vs modern M_{orb} vs the anticommutator current $\frac{1}{2}\{v_x, L_z\}$)?

Q5. Is the texture-induced $\sigma_{xy}^{L_z}$ converged with respect to lattice size L and k-mesh, and does the FM-subtracted signal remain stable as $L \to \infty$ at fixed λ/L?

Verdict

PARTIAL (headline REPLICATED). The paper's central, novel claim — a finite topological orbital Hall effect for s-electrons *without* SOC, purely texture-driven, with orbital response dominating spin by $\sim$3 orders of magnitude — reproduces cleanly, independently, and robustly (C1 ✓, C3-ordering ✓). The precise *area-scaling exponent* (C2) and *integer charge quantization* (C3-charge) are only partially reproduced and require the paper's modern orbital-magnetization operator and a skyrmion-lattice supercell, respectively — ingredients out of scope for this minutes-scale CPU run. No claim numbers were fabricated; where fidelity fell short it is reported as PARTIAL with the mechanistic reason.

Artifacts: `work/reproduce.py`, `work/results.json`, `work/COMPUTE_NOTES.md`, `work/figs/scaling.png`. Runtime $\approx$4 min, pure `numpy`, CPU. Paper data: DOI:10.5281/zenodo.13919926.